

Anshul Sanghi

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Examination	University	Institute	Year	CPI/%
Post Graduation (M.Tech in Computer Science)	IIT Bombay	IIT Bombay	2024	8.81
Graduation (B.Tech in Computer Science)	IIIT Gwalior	IIIT Gwalior	2022	8.42
Intermediate/+2	CBSE	Delhi Public School, Durg	2018	92.4

WORK EXPERIENCE

- Software Engineer | Microsoft Corporation**
Team: Customer Experience Division, Microsoft Security (Jul '24 - Current)
 - Spearheaded the development of Prompt Playground, MCP Server, and Copilot Agent, driving 7,500+ AI insights API hits
 - Delivered full-stack solutions by building scalable .NET backend APIs and React-based visual dashboards
 - Migrated portal from Synapse DB to Kusto DB, reducing API response times by 50% and lowering failure rates below 10%
 - Built context pipeline for NL2KQL agent and expanded prompt library to 20+ scenarios, enhancing agent response quality
 - Improved app insights and caching mechanisms, enabling personalized experiences and improved platform efficiency
 - Collaborated with cross-functional teams to develop AI-generated customer reports, giving PMs a one-stop data summary
- Software Developer Trainee | One97 Communications Ltd. (Paytm)**
Team: Mapping service and merchant integration (Payment Gateway) (Jan '22 - May '22)
 - Implemented feature flags for switching database caching mechanisms in mapping service APIs
 - Collaborated with cross-functional teams and units to identify and optimize feature flags in the codebase
 - Underwent comprehensive training in Java, Spring boot, Jenkins, Kibana, and canary server deployment
- Analytics Valley Technologies Pvt. Ltd.**
 - SDE Intern | Research and Development (Nissan R&D client)** (May '21 - Jul '21)
 - Built a Mask R-CNN app for real-time nanoparticle image segmentation, achieving nearly 80% initial accuracy
 - Developed a CNN U-Net denoiser for photo-emission spectra, improving impurity detection accuracy by 15%
 - Python Intern | Development and testing (Suburban diagnostics client)** (May '19 - Aug '19)
 - Developed a push notification service, integrating thread pools to manage multi-level and multi-priority queues
 - Deployed the service for android to handle dependency delay management for day to day CRM operations

COLLABORATIVE AI INNOVATION INITIATIVE

- Agentic Movie Generation Pipeline** (Apr '26 - Aug '26)
 - Architected a LangGraph-based agentic pipeline that transforms screenplays into cinema-quality generated videos
 - Automated character and location creation, shot decomposition, and camera planning across cinematic styles
 - Developed an observability-driven prompt optimization workflow using live pipeline telemetry to improve generation quality
 - Built an evaluation layer with self-validation and regeneration loops to improve consistency and output quality

TECHNICAL SKILLS

Languages & Query Languages	C#, Python, C++, C, Bash, SQL, KQL
Agentic AI & Machine Learning	LangGraph, AutoGen, MCP, Azure AI SDK, Prompt Engineering, LLM Evals
Backend & Frontend Development	ASP.NET Core, React, REST APIs, Flask, Selenium
Cloud, Containers & Observability	Azure (Infra and Services), AWS, Kubernetes, Docker, Langfuse, Linux, Git

ACADEMIC ACHIEVEMENTS

- Winner** of Microsoft Security Hackathon for building AI Generated Customer Reports 2025
- All India Rank 10 and 206**, in Graduate Aptitude Test in Engineering (GATE) among 100k+ candidates 2021-22
- Global ranks 85, 203, and 332**, in International Mathematics Olympiad (SOF) 2014-17
- Global rank 267 and All India Rank 17**, in Google Hashcode among 10000+ teams. 2022
- Chhattisgarh State Topper**, in National Science Talent Search Examination (NSTSE) 2017
- Codechef rating of 1887**, achieved through securing **Global ranks of 44, 171, 291, 322 and 357** 2020-22
- All India Rank 11021**, in Joint Entrance Examination(JEE) Mains among 1 Million+ candidates 2018
- Leaderboard topper**, title received for outperforming peers in club-hosted events throughout the year 2019

CERTIFICATIONS

- Azure AI Fundamentals**, Microsoft certification for excellence in Azure AI ecosystem. 2025
- Machine Learning Foundations**, University of Washington for excellence in machine learning basics. 2020

RESEARCH & ACADEMIC CONTRIBUTIONS

- **M.Tech Thesis: Advancing GPU Resource Management and Virtualization in Kubernetes** (Jun '23 - Jun '24)
 - Built K8 device plugin to enhance resource allocation flexibility via independent GPU memory and compute control
 - Profiled HPC, Math, and ML workloads to uncover distinct configurations delivering equivalent performance
 - Demonstrated the efficacy of the plugin with a suite of HPC workloads and real life use-cases
 - Extended Kubernetes scheduler to filter pertinent configurations, for improved compute & memory utilization
- **M.Tech Seminar: GPU Multiplexing and Virtualization Solutions for Cloud** (Jan '23 - May '23)
 - Conducted an in-depth analysis of NVIDIA's software and hardware-assisted GPU virtualization solutions
 - Researched open-source software-assisted techniques to enhance features and refine multiplexing control
 - Comprehensive study of scheduling, execution, and management policies for GPU workloads in cloud setup
- **B.Tech Project: Healthcare assistant for medical disease diagnosis** (Jan '22 - May '22)
 - Engineered a medical diagnostic chatbot using a knowledge graph and series of text similarity models
 - Leveraged disease-based attributes like symptoms and age group to effectively filter and identify diseases
 - Developed a user-friendly Android application to enhance accessibility and user experience

PERSONAL AGENTIC AI PROJECTS

- **Multi-Agent Market Intelligence & Trading Platform** (Mar '26 - Apr '26)
 - Built persona-driven LangGraph agents mirroring hedge fund roles to debate market signals and make trading decisions
 - Engineered long-term and intraday workflows unifying market data, news, NSE/BSE forums, and Reddit intelligence
 - Developed an interactive UI tracing agent discussions, decision rationale, and end-to-end trade execution

ACADEMIC PROJECTS & CAPSTONE WORK

- **Lightweight Containerization Framework using Linux Primitives** (Virtualization and Cloud Computing, Spring 2023)
 - Implemented a containerized environment using cgroups, namespaces and chroot to mirror Docker's functionality
 - Successfully ensured robust resource and process isolation between the generated environments
- **Lightning Network on Top of Ethereum** (Blockchain, Cryptocurrency and smart contracts, Spring 2023)
 - Implemented a lightning network through smart contracts on top of Ethereum using Solidity
 - Conducted practical simulations of adversarial behaviours with focus on selfish and stubborn mining attacks
- **Stock Price Prediction using Stock Correlation** (Foundations of Machine Learning, Autumn 2022)
 - Assessed the impact of cross-sector correlation-based generalized features on LSTM model performance
 - Compared results with a specialized LSTM model trained exclusively on the target stock's data
- **Honeypot System For Intruder Detection** (Informations system security, Autumn 2019)
 - Created a network utility to trap attackers within a virtual file system inside a docker container
 - Employed an enticing vulnerability in the telnet port to lure and analyze malicious activities effectively

PRIMARY ACADEMIC COURSES

Design and Engineering of Computing Systems | AA
Topics in Virtualization and Cloud Computing | AB
Foundations of Machine Learning | AB

Seminar in GPU virtualization techniques | AA
Blockchains, Cryptocurrencies and Smart contracts | AB
Algorithms and Complexity | AB

POSITION OF RESPONSIBILITIES

- **AI Instructor** for Microsoft's 2026 fresher cohort on leveraging AI across the security domain (Jul 2026)
- **Lead Trainer** for the CXE team offsite on leveraging AI for customer interactions (Nov 2025)
- **Cloud Computing Instructor** for a 2-day workshop attended by 40+ students at RV College, Bangalore (Jan 2025)
- **Teaching Assistantship**
 - CS 347, 333: **Operating Systems Theory + Lab** under Prof. Purushottam Kulkarni (Aug '23 - Jun '24)
 - CS 104: **Software Systems Lab** under Prof. Kameswari Chebrolu (Spring 2023)
 - CS 101: **Computer Programming and Utilization** under Prof. Ajit Rajwade (Autumn 2022)
- **Mentored** 15+ juniors in building intra-network torrent as winter project (Dec '20 - Jan '21)
- **Volunteered** in design and marketing team for annual techno-managerial fest Infotsav (Nov '18 - Feb '19)

HOBBIES AND INTERESTS

- Possess **basic proficiency in French**, acquired through three years of formal study as third language
- Showcased basketball prowess by participating in **Sub-junior State Basketball Championship** in 2012 and 2013
- Avid **sketch artist** with a passion for drawing, painting and origami cultivated from a very early age
- **Educator** on Superprof, mentoring 15+ students to date in programming, AI tools, and other technical subjects